

ShadowTrace AI — Behavioral Cybercrime Intelligence for Coordinated Identity Detection

OVERVIEW

Cybercrime actors frequently operate multiple aliases across social media platforms to evade detection and impersonate legitimate services. Manually correlating these accounts is slow, inconsistent, and difficult to sustain across active investigations. Unlike conventional keyword or metadata-matching tools, ShadowTrace AI operates on behavioral patterns — making it resilient to alias changes and account recycling. The system is a SOCMINT prototype that combines behavioral fingerprinting and Explainable AI to identify coordinated digital identities from publicly available data.

TECHNICAL APPROACH

The system extracts five behavioral signals across submitted handles — posting timing patterns, multilingual phrase similarity (Hindi + English fraud templates), hashtag cluster overlap, stylometric writing analysis, and repeated payment link patterns. Similarity scoring uses multilingual sentence-transformer embeddings. A weighted explainable scoring engine produces a 0–100% Coordinated Activity Score, classified into three actionable tiers: HIGH (escalate), MEDIUM (monitor), and LOW (archive).

INVESTIGATOR OUTPUT

Each analysis generates a plain-language XAI reasoning summary explaining which signals triggered the verdict. A timestamped incident report is produced with legal references (IT Act §66C, §66D; IPC §420), relationship graph visualisation, and a recommended action — formatted for direct handoff to a cybercrime unit.

COMPLIANCE & VALIDATION

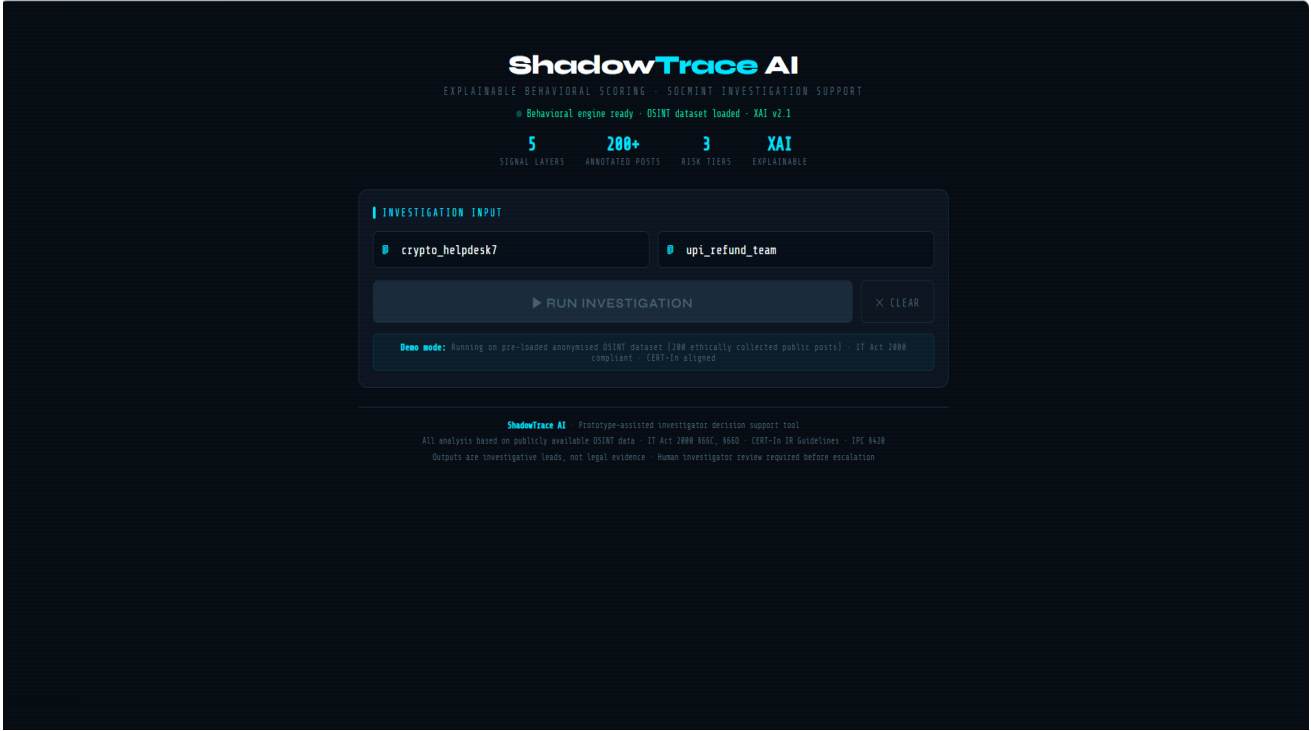
The system was evaluated on a curated dataset of 200 anonymised, ethically collected public posts across manually labelled account pairs. Behavioral signal extraction and similarity scoring were validated through a structured test protocol comparing system verdicts against ground-truth annotations. All analysis operates exclusively on publicly available OSINT data, aligned with CERT-In Incident Response Guidelines and IT Act 2000. The scoring pipeline is deterministic, ensuring reproducibility across repeated analyses. A functional end-to-end prototype is complete and available for live demonstration.

EXAMPLE BEHAVIORAL INDICATORS

Signal	Example
Phrase Match	Repeated scam phrases such as 'refund pending' and 'verify now'
Timing Pattern	Accounts posting within identical 5-minute windows

Stylometry	Consistent punctuation, emoji usage, and writing structure
Payment Links	Shared shortened payment URLs detected across handles

SYSTEM INTERFACE — LIVE PROTOTYPE



ShadowTrace AI live investigator interface showing behavioral investigation workflow, OSINT dataset analysis, and explainable behavioral scoring engine.

DEPLOYMENT & DEMONSTRATION

The prototype is designed as a deployable investigator-support workflow with interactive analysis, relationship graph visualisation, deterministic behavioral scoring, and downloadable incident report generation. The system is intended for demonstration in cybercrime investigation and SOCMINT evaluation environments.

***ShadowTrace AI** · Prototype-assisted investigator decision support tool*
All analysis based on publicly available OSINT data · CERT-In aligned · IT Act 2000 compliant
Outputs are investigative leads, not legal evidence · Human investigator review required before escalation